SplatAtlas: Benchmarking Photometric Saturation and Representational Divergence in 3D Gaussian Splatting

Author Name
Department / Institution
email@domain.com

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Abstract

3D Gaussian Splatting has reached a regime in which standard photometric metrics offer compressed method-level separation: across 31 methods evaluated under a unified protocol on 31 scenes, the cross-method per-scene PSNR interquartile range (IQR) has a median of only 0.586 dB (bootstrap 95% CI $[0.462, 0.672]$ dB), with a cohort-median seed standard deviation of 0.058 dB (see section 3.3). We introduce SplatAtlas, a uniformly orchestrated benchmark covering 31 methods × 31 scenes drawn from four standard datasets, with complete training trajectories and end-state Gaussian fields recorded per run. A two-way ANOVA on this corpus reveals an asymmetric variance attribution: PSNR is scene-dominated ($> 98\%$) while opacity and anisotropy are method-dominated (86%, 68%); scale is jointly method- and scene-dependent (43% vs. 45%). We refer to this pattern—methods reaching near-identical photometric outputs while allocating their internal Gaussian primitives in markedly different ways—as *representational divergence*. Two candidate organising principles for the resulting method-dominated divergence find no support in the data, and a follow-up check on SH–geometry compensation finds that the cell-level signature runs counter to the simplest prediction. We further develop pixel-level counterfactual profiling, a post-hoc diagnostic derived from the rendering equation, currently demonstrated on the 9 methods using standard 3DGS primitives (279-cell sweep in under one GPU-hour); within the saturated cluster its channels exhibit relative IQR (defined in section 5.4) up to 22.8× that of PSNR in the dB domain (3.7× in the MSE domain; both from opacity pathology rate), and two of four paired comparisons against vanilla reach unanimous direction agreement with their stated design intent.

1 Introduction

We approach 3DGS Kerbl et al. [2023] evaluation as a measurement study: the population of methods trained under a unified protocol is treated as the analytic object, rather than as candidates for a leaderboard. Under this framing, output-level and representation-level metrics carry distinct information once PSNR saturates—the former offer limited separation within the saturated cluster, while the latter retain method-level structure. We refer to the resulting gap—methods that converge to nearly identical photometric outputs but allocate their internal Gaussian primitives in markedly different ways—as *representational divergence*, and treat it as the central object of study. This paper operationalises that position—not as a settled matter—through the unified reproduction protocol of section 3 and the counterfactual profiling tool of section 5.

C1: A unified-protocol corpus of 31 methods × 31 scenes with full trajectories. SplatAtlas covers 31 methods × 31 scenes across four datasets under a unified reproduction protocol (section 3); the corpus, profiling tool, audit logs, and Croissant Akhtar et al. [2024] metadata are released as a single artifact bundle. **C2: Photometric output saturates while internal**

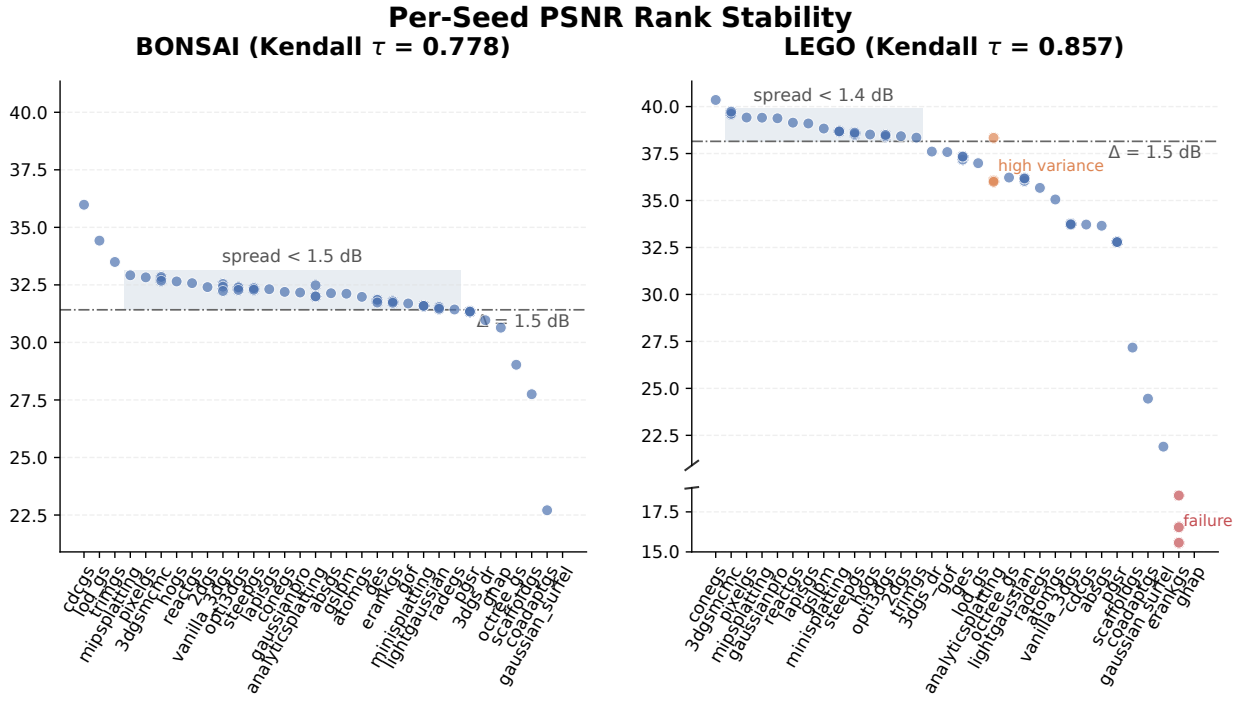


Figure 1: **The saturation puzzle.** Per-seed PSNR rankings on BONSAI (Kendall $\tau = 0.778$, spread < 1.5 dB in the top cluster) and LEGO (Kendall $\tau = 0.857$, spread < 1.4 dB). The red dot marks the failure cell within the seed-sweep (`erankgs` × LEGO ≈ 17 dB across all seeds; full NeRF-Synthetic failure profile in section 3.3 and appendix D).

representation diverges. A two-way ANOVA on end-state attributes reveals an asymmetric variance attribution: PSNR is scene-dominated ($> 98\%$), whereas opacity and anisotropy are method-dominated (86%, 68%) and scale is jointly method- and scene-dependent (43% vs. 45%; section 4.2). Two candidate organising principles for the resulting method-dominated divergence find no support in the data—stable feedforward dynamics across representational dimensions (section 4.3; no dimension-pair lead–lag direction across 28 methods, all $p > 0.05$) and correspondence with the literature taxonomy (section 4.4; ARI $= 0.025$ between data-driven trajectory clusters and the five-category labelling). A follow-up check on SH–geometry compensation finds that the cell-level signature runs counter to the simplest prediction (appendix F; $r = 0.417$ positive rather than the predicted negative, $n = 868$). **C3: Counterfactual profiling recovers method-level distinctions invisible to PSNR.** A pixel-level counterfactual profiling tool probes the channel-specific sensitivity of individual representation components on a trained Gaussian field; the full 279-cell sweep completes in under one GPU-hour. Within the saturated cluster, counterfactual channels exhibit relative IQR up to 22.8× that of PSNR in the dB domain (3.7× in the MSE domain) (section 5.4); 2/4 paired comparisons against vanilla reach direction agreement of 1.000 with their stated design intent and a third reaches 0.839, while the fourth yields a consistent counter-finding for LightGaussian pruning (section 5.6).

Throughout this paper, *cell* refers to a single (method, scene) combination; *cell-level* analysis operates on per-cell aggregate statistics, while *pixel-level* analysis operates on individual pixels of rendered test views.



Figure 2: **Illustrative saturated group on TRUCK (Tanks & Temples).** Five methods reach near-identical PSNR (25.28–25.80 dB) while exhibiting markedly different anisotropy (highlighted; range 6.7–33,778). Output saturation coexists with representation divergence.

2 Background and Related Work

Methods built on 3D Gaussian Splatting are commonly grouped by design intent. We adopt a five-category labelling—FREE, GEO_REG, PRIM_CTRL, RENDER, and COMPRESS—adapted from the taxonomy of Bao et al. [2025], with category boundaries adjusted to fit the $N_2 = 31$ methods evaluated in this work; the labels reflect the design motivation behind each method, such as regularising the Gaussian field toward a surface, modifying the densification policy, restructuring the rasterisation pipeline, or compressing a trained field. Per-method labels for the $N_2 = 31$ methods evaluated in this work are listed in table 6. This grouping reflects design philosophy rather than dynamical behaviour; section 4.4 reports that data-driven clusters over training trajectories are nearly orthogonal to this literature taxonomy.

SplatAtlas is positioned in the measurement-study tradition: rather than proposing another 3DGS variant or treating the method set as a leaderboard, we use a unified reproduction protocol to ask what standard photometric metrics still distinguish, what they fail to distinguish, and which representation-level measurements remain informative after photometric saturation. PSNR, SSIM, and LPIPS remain useful output-level references, while the trajectory and counterfactual analyses developed here probe the method-level structure that those scalars compress away.

3 SplatAtlas: Benchmark and Protocol

3.1 Reproduction Protocol

All $N_2 = 31$ methods are trained under a four-component protocol: per-method Conda environments isolate dependencies (retaining each method's original stack), a Python wrapper normalises entry-points to a common interface, YAML configurations declare hyperparameters and seeds per (method, scene), and audit logs record run metadata (git commit, hyperparameters, hardware). No internal algorithm is modified; the protocol unifies only the external interface and metric-injection points. Where a method's own code includes fallback paths (e.g. random initialisation when no SfM point cloud is available), these execute as shipped; no benchmark-side adaptation is applied.

3.2 Inclusion Tiers and Datasets

Table 1 defines two analytic tiers. $N_2 = 31$ methods support the trajectory and end-state analyses of section 4; $N_3 = 9$ supports the counterfactual profiling of section 5, restricted to methods using

Table 1: Inclusion tiers. Full method inventory in table 6.

Tier	N	Inclusion criterion	Used in
N_2	31	Complete trajectory & end-state PLY	section 4
N_3	9	$N_2 \cap$ standard 3DGS primitives	section 5

standard 3DGS primitives—anchor-based and surfel-based variants are excluded because the SH-zero and opacity-clamp operators are not comparably defined on non-standard primitives.

Evaluation spans 31 scenes from four standard benchmarks: Tanks&Temples Knapitsch et al. [2017], Deep Blending Hedman et al. [2018], Mip-NeRF 360 Barron et al. [2022], and NeRF-Synthetic Mildenhall et al. [2020]. For COLMAP-style real-world scenes, we follow the standard 3DGS hold-every-eighth-frame protocol; for NeRF-Synthetic, we use the local train/test/validation splits reported in Appendix B. All scenes are evaluated at the resolutions specified in Appendix B. The full method inventory (with category labels and citations) is listed in table 6; per-scene properties are listed in Appendix B.

Trajectory analyses (sections 4.3 and 4.4 and Appendix F) further exclude methods whose primitive representation is not directly comparable on the four physical dimensions (opacity, scale, anisotropy, SH) tracked by the trajectory features; cross-method counterfactual comparisons (sections 5.4 and 5.5) additionally exclude AnalyticSplatting on rasteriser-backend grounds (section 5.8).

3.3 Hyperparameter Policy and Seed Variance

All methods use each codebase's default hyperparameters, including training iterations (30k for all N_3 methods; two N_2 methods train for longer or shorter budgets under their own defaults). A seed sweep over two scenes—BONSAI (Mip-NeRF 360) and LEGO (NeRF-Synthetic)—with three seeds per method, on the 10 methods of N_2 for which seed runs completed (full per-cell summary in Appendix D), establishes a per-cell PSNR standard deviation of cohort median 0.058 dB and $P_{95} = 0.384$ dB; we report this P_{95} value as a conservative empirical reference scale for seed-induced variation. The P_{95} of 0.384 dB is dominated by a single high-variance cell (AnalyticSplatting $\times$ LEGO, $\sigma = 1.333$ dB; Appendix D). These figures characterise within-cell seed noise and are not formal significance thresholds for between-method comparisons. Kendall's τ over per-seed PSNR rankings is 0.778 on BONSAI ($n = 10$) and 0.857 on LEGO ($n = 8$; excluding `erankgs` for training failure and `vanilla_3dgs` for missing seed data at analysis time). Two cells deviate notably: `analyticsplatting` $\times$ LEGO exhibits anomalously high seed sensitivity ($\sigma = 1.33$ dB, range 2.34 dB; all three reconstructions valid) and carries an outlier annotation in PSNR-absolute statistics throughout; `erankgs` $\times$ LEGO fails entirely (PSNR ≈ 17 dB across all seeds) and is excluded from seed-variance statistics but retained in benchmark tables as a failure-mode exemplar. Additional method-level training anomalies are documented in Appendix E.1.

3.4 Compute and Release

Runs were executed on AutoDL cloud instances with NVIDIA RTX 4080 SUPER GPUs. The benchmark is released as two HuggingFace bundles under CC-BY-4.0 (`https://huggingface.co/datasets/KCBtheone`; the link is anonymised for review and will be transferred to the authors' identifiable account upon acceptance): a core bundle (aggregated CSVs, audit logs, Croissant metadata, README) and a runs bundle (per-experiment trajectories, end-state fields, checkpoints at 5k/10k/20k/30k). Adding a new method requires roughly 300 lines of wrapper code plus one YAML entry.

4 Empirical Findings

4.1 Photometric Saturation

Across $N_2 = 31$ methods, the cross-method per-scene PSNR IQR has a median of 0.586 dB (bootstrap 95 % CI $[0.462, 0.672]$ dB), with corresponding values of 0.009 for SSIM Wang et al. [2004] (CI $[0.006, 0.012]$) and 0.018 for LPIPS Zhang et al. [2018] (CI $[0.013, 0.022]$). PSNR, SSIM, and LPIPS are nearly interchangeable as scalar summaries within the saturated regime (PCA in Appendix I: PC1 = 88.4%, near-equal loadings).[1]

4.2 Method-Dominated Representation

While photometric output is almost entirely a function of the scene, internal Gaussian-field attributes are largely a function of the method. The two views give an asymmetric picture of the same trained models.

A two-way ANOVA on end-state attributes (method × scene, no interaction[2]) yields $\mathrm{SS}_{\mathrm{method}}/\mathrm{SS}_{\mathrm{total}} = 0.857$ (CI $[0.835, 0.888]$) for opacity and 0.677 (CI $[0.603, 0.776]$) for anisotropy. Scale occupies an intermediate position: method accounts for 0.426 (CI $[0.342, 0.535]$) and scene for 0.451, making it jointly method- and scene-dependent rather than method-dominated. The same decomposition applied to PSNR reverses the pattern: $\mathrm{SS}_{\mathrm{scene}}/\mathrm{SS}_{\mathrm{total}} = 0.986$ (CI $[0.984, 0.989]$), with method accounting for only 1.4%. The photometric loss strongly constrains the output while leaving the internal representation largely free to vary across methods—an asymmetric variance attribution that frames the rest of this section (fig. 3).

Robustness caveat. Approximately 8% of cells exhibit method-level training anomalies (Appendix E.1). These cells are retained in the ANOVA to reflect actual protocol-level behaviour, but extreme failures (e.g. PSNR < 10 dB) may inflate $\mathrm{SS}_{\mathrm{method}}$. The reported method-attributed variance shares for opacity and anisotropy may therefore be inflated by these cells; the qualitative pattern—opacity and anisotropy method-dominated, PSNR scene-dominated—is expected to be robust to exclusion of failure cells, but a formal sensitivity analysis across exclusion criteria is deferred to future work.

The no-interaction model does not separately identify the method × scene interaction: the residual term absorbs both interaction and within-cell noise. The extreme asymmetry ($\mathrm{SS}_{\mathrm{method}}/\mathrm{SS}_{\mathrm{total}} = 0.857$ for opacity vs. $\mathrm{SS}_{\mathrm{scene}}/\mathrm{SS}_{\mathrm{total}} = 0.986$ for PSNR) makes qualitative reversal of the attribution implausible, but does not formally exclude interaction effects of unknown magnitude. Estimating the interaction term directly would require replicated observations per cell.

The exception is `sh_energy_ratio` (scene 57%, method 17%), behaving as scene-dominated. This is consistent with SH energy tracking scene difficulty (proxied by training loss); a cell-level test of the simplest SH–geometry compensation hypothesis ($n = 868$) yields the opposite-sign signature (Spearman $r = 0.417$; Appendix F).

4.3 No Stable Feedforward Dynamics

A second candidate is that one representational dimension drives the others, producing a stable feedforward order across training. We test for such ordering through cross-correlation analysis on the

[1] Trajectory and end-state distance matrices correspond strongly (Mantel $r = 0.820$ globally; per-dimension 0.625–0.925), supporting trajectory-level analysis as an end-state proxy in sections 4.3 and 4.4.

[2] Each (method, scene) cell contains a single observation; under this design the saturated method × scene interaction model has zero residual degrees of freedom. We therefore report the no-interaction model, in which the $(N_{\mathrm{methods}} - 1)(N_{\mathrm{scenes}} - 1)$ interaction sum of squares serves as the error term—the standard treatment for one-observation-per-cell two-way ANOVA. Under this model the residual confounds method × scene interaction with within-cell stochastic variation; the two are not separately identifiable without replicated observations.

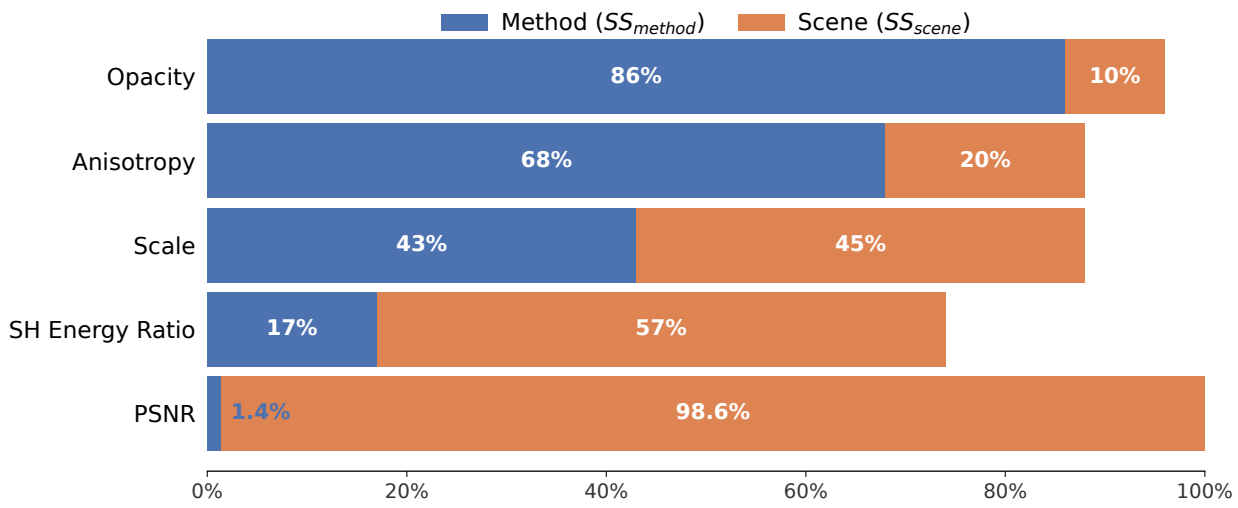


Figure 3: Asymmetric variance attribution across end-state attributes and PSNR. Each bar pair shows the method-attributed (blue) and scene-attributed (orange) share of total sum of squares from a two-way ANOVA (method × scene, no interaction). Opacity and anisotropy are method-dominated (86%, 68%); scale is jointly method- and scene-dependent (method 43%, scene 45%); PSNR is scene-dominated (98.6%). The `sh_energy_ratio` exception (57% scene) is consistent with SH energy tracking scene difficulty (proxied by training loss) rather than method-specific design (Appendix F). Bar pairs for the end-state attributes do not sum to 100%; the gap is the residual term of the one-observation-per-cell no-interaction ANOVA model (footnote 2).

four physical dimensions, supplemented by three dynamical observations on densification, freezing, and adaptability. None supports a directed dynamical structure.

Across all 28 methods and six dimension pairs, no pair rejects the null of no dominant lead–lag direction (binomial test against H_0: $p = 0.5$, all $p > 0.05$); consistency values range 0.430–0.553 (median 0.481) and median peak lags lie within 0–1.0 steps $\times 10^3$ (fig. 4A). The four dimensions co-vary in parallel rather than in sequence.

Densification mechanisms perturb representational dimensions non-uniformly, and late-stage structural freezing is widespread (Appendix J; fig. 4 panels B–C).

4.4 Literature Taxonomy Misaligned

A third candidate is that the literature taxonomy of 3DGS methods—FREE / GEO_REG / PRIM_CTRL / RENDER / COMPRESS—captures functionally distinct dynamical behaviours. We test this by comparing data-driven trajectory clusters against literature labels: with $k = 4$ k-means clustering on standardised trajectory features, ARI = 0.025 and NMI = 0.061 against the five literature categories; 17 of 28 methods are reclassified (Appendix K). Literature category should not be used as a surrogate for trajectory-level behaviour in downstream method comparisons.

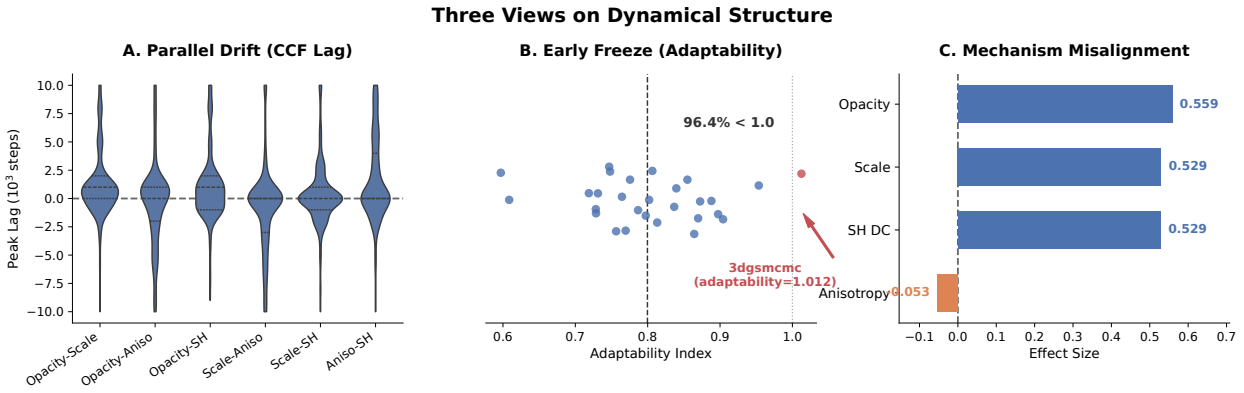


Figure 4: Three views on dynamical structure. **(A)** Per-pair cross-correlation peak lag distributions across 28 methods (units: 10^3 optimisation steps). All six pairs centred near zero with broad support. **(B)** Adaptability index across 28 methods; median 0.800, 96.4% below 1.0. **(C)** Densification-window effect sizes on the four physical dimensions; opacity / scale / *sh_dc_mag* show consistent medium-to-large effects, anisotropy near-zero with opposite sign. None of the three views supports a directed dynamical structure across methods.

5 Pixel-Level Counterfactual Profiling

5.1 Operator Derivation from the Rendering Equation

The 3DGS rendering equation alpha-composites a sequence of Gaussian primitives over a background:

$$C(p) \;=\; \sum_{i=1}^{N} c_i(\mathbf{d})\, w_i \;+\; T_{N+1}\, C^{\mathrm{bg}}, \qquad w_i = \alpha_i T_i, \quad \alpha_i = o_i\, G_i(p), \tag{1}$$

with $T_i = \prod_{j<i}(1-\alpha_j)$. Each primitive carries four learnable parameter groups: position μ_i, covariance Σ_i (scale and rotation), opacity o_i, and view-dependent appearance f_i (SH coefficients). A counterfactual operator must correspond to placing one of these groups in a well-defined neutral state.

Two of the four groups admit a natural neutral state. For f_i, zeroing the AC SH coefficients ($\ell \geq 1$) while preserving the DC term yields an SH-removed render with no view dependence. For o_i, clamping all opacities to one yields a *unit-opacity-amplitude* render: each primitive's peak contribution is maximised, but the Gaussian spatial falloff $G_i(p)$ still modulates α_i, so the render is not strictly fully opaque. We label the resulting deltas in per-pixel error $\Delta\varepsilon_{\mathrm{SH}}$ and $\Delta\varepsilon_{\mathrm{opa}}$.

The geometric groups (μ_i, Σ_i) admit no comparably natural neutral state: zeroing positions is undefined under the projective geometry of eq. (1), and clamping scales breaks the covariance structure that the equation depends on. We therefore do not intervene on geometry directly; instead we *read off* a structural complement of the rendering equation, the transmittance term T_{N+1}, which quantifies the share of pixel energy that the Gaussian field fails to cover and must be supplied by C^{bg}. The corresponding channel ϕ_{cov} is a measurement, not an intervention: it reads off the transmittance-weighted background exposure from the rendering equation rather than substituting a parameter. Because the L_1 norm is not additively decomposable across foreground and background terms, ϕ_{cov} quantifies background exposure rather than an exact error share. On pixels that are fully covered ($T_{N+1} < 0.05$) and where both the SH and opacity operators improve accuracy, the remaining per-pixel error is attributed to the residual channel $\varepsilon_{\mathrm{res}}$. The residual is a catch-all by construction and includes geometric error along with any other unattributed terms; we report it descriptively rather than as a measurement of geometric error.

Table 2: Counterfactual channel definitions. Net-effect channels measure the average change in per-pixel L_1 error under the operator; rate channels measure the fraction of pixels for which the operator *reduces* error, indicating representational pathology.

Channel	Operation	Symbol	Physical meaning
SH net effect	zero all AC SH ($\ell \geq 1$)	$\Delta\varepsilon_{\text{SH}}$	view-dependent colour contribution
SH corruption rate	fraction where zeroing SH reduces error	ρ_{SH}	SH over-fit pixel share
Opacity net effect	set all opacities to 1	$\Delta\varepsilon_{\text{opa}}$	sensitivity to sub-unit opacity amplitud
Opacity pathology rate	fraction where clamping reduces error	ρ_{opa}	fraction of pixels where learned opacity
Coverage error fraction	transmittance-weighted bg error / total error	ϕ_{cov}	background-transmittance exposure ind
Residual error	error on fully-covered, SH/opa-healthy pixels	ε_{res}	remainder unexplained by the above

5.2 Operational Framework

Two trained models can reach indistinguishable PSNR on the same held-out views while differing in how their internal representations allocate error. We define a *counterfactual delta* as the change in per-pixel error when one channel of section 5.1 is altered or read off, with all other components held fixed. This is operational rather than causal: we describe intervention effects under a fixed rendering model, not structural causation. For each trained model and test view we render three tuples—the *full* model, an *SH-zeroed* variant (f_i AC coefficients set to zero), and an *opacity-clamped* variant ($o_i \leftarrow 1$)—and compute per-pixel L_1 residuals against the ground-truth image. Coverage is read off from the transmittance maps of the full render. We aggregate across views by mean and across scenes by median, robustifying against pathological cells (e.g. $\phi_{\text{cov}} = 0.74$ on FICUS). All renderings use a single unified backend (`gsplat` Ye et al. [2025]); rank-order consistency with native rasterisers is verified in table 5 (median Spearman $\rho = 1.000$ across 19 scenes). Three preconditions support delta interpretability: rendering is deterministic ($\varepsilon_\tau = 0$); joint interventions are approximately additive (interaction residual -0.7%); and a single backend serves all methods.

5.3 Counterfactual Channels

Table 2 lists the six channels. Channels answer separate counterfactual questions ("what happens if X is removed?") rather than partition total error orthogonally; they are not statistically independent, since all operate on the same set of trained Gaussians. The residual channel is a catch-all by construction and is reported descriptively, not interpreted as a measurement of geometric error.

5.4 Saturation-Zone Divergence

The saturated cluster comprises the largest method subset within the N_3-comparable group (8 methods) whose cross-scene median PSNR (under the gsplat backend) falls within approximately 1.5 dB of the group's top performer, yielding $k = 5$ methods (Appendix L). We quantify within-cluster method-level dispersion via the *relative IQR*:

$$\text{relIQR}(x) \;=\; \frac{Q_3(x) - Q_1(x)}{|\text{median}(x)|}, \tag{2}$$

computed over the k method-level cross-scene medians. The ratio $R_c = \text{relIQR}(c) \,/\, \text{relIQR}(\text{PSNR})$ measures how much more (or less) a counterfactual channel c disperses across methods relative to PSNR. Table 3 reports R_c in both the dB domain (PSNR as reported) and the MSE domain (PSNR converted via $\text{MSE} = 10^{-\text{PSNR}/10}$). In the dB domain, five of six channels yield $R_c > 1$: opacity pathology rate is highest ($R_c = 22.8\times$), followed by opacity net effect ($14.8\times$), coverage error fraction ($10.4\times$), SH net effect ($6.9\times$), and residual error ($1.7\times$). SH corruption rate ($R_c = 0.95\times$) shows no additional dispersion beyond PSNR. Because PSNR is a logarithmic scale, R_c is not scale-invariant: in the MSE domain all ratios compress, with the highest being opacity pathology rate at $3.7\times$. The

qualitative finding—that multiple counterfactual channels disperse more than PSNR within the saturated cluster—holds in both domains, though the magnitudes differ substantially.

Additionally, R_c is conditional on the PSNR-based cluster definition: restricting to methods within $\approx 1.5\,\mathrm{dB}$ compresses the PSNR denominator, mechanically inflating the ratio. The MSE-domain values in table 3 and the wider-subset analysis (Appendix H) provide context for this effect.

Table 3: Relative IQR ratio R_c within the saturated cluster ($k = 5$). Values above 1× indicate greater method-level dispersion than PSNR.

Channel	relIQR	R_c (dB)	R_c (MSE)
Opacity pathology rate	0.995	22.8×	3.7×
Opacity net effect	0.645	14.8×	2.4×
Coverage error fraction	0.453	10.4×	1.7×
SH net effect	0.302	6.9×	1.1×
Residual error	0.072	1.7×	0.3×
SH corruption rate	0.041	0.95×	0.15×
PSNR	0.044	1.0× (ref)	1.0× (ref)

Sensitivity on the wider N_3-comparable subset (8 methods) preserves the qualitative finding, although the exact channel ordering differs (Appendix H).

5.5 Cross-Method Patterns

Figure 5 displays each method's cross-scene median profile (PSNR + 6 diagnostic channels + raw coverage as context). Two channels exhibit pronounced cross-method structure: on ϕ_{cov}, eRankGS Hyung et al. [2024] reaches 0.002 vs. vanilla-3DGS 0.120 (60× gap); on $\Delta\varepsilon_{\mathrm{opa}}$, PGSR Chen et al. [2025a] reaches 0.026 vs. 0.130 (5× gap)—both outlier directions match design intent. $\Delta\varepsilon_{\mathrm{SH}}$ is compressed within 0.007–0.017 across the 8 comparable methods, indicating limited cross-method discriminative power on SH.

Within the 8 comparable methods, the two opacity subchannels can move in opposite directions under the same constraint—PGSR has the *lowest* $\Delta\varepsilon_{\mathrm{opa}}$ (0.026) yet the *highest* ρ_{opa} (0.274 vs. 0.077 for vanilla-3DGS), a pattern invisible to PSNR.

Six-channel separation profile. To test whether PGSR's divergence is confined to the directly tested channel or extends across the full diagnostic profile, we compute Wilcoxon signed-rank tests (paired by scene) for PGSR against each of the 7 comparable N_3 methods, per channel, with Holm correction across all 42 tests (7 methods × 6 channels). PGSR separates strongly on the opacity subsystem: $\Delta\varepsilon_{\mathrm{opa}}$ (median 0.026 vs. 0.086, significant in 7/7 pairings after correction) and ρ_{opa} (0.274 vs. 0.116, 6/7). $\Delta\varepsilon_{\mathrm{SH}}$ reaches significance in 6/7 pairings, though with compressed absolute magnitude (range 0.007–0.017 across all 8 methods). ρ_{SH} is significant in 3/7 and ϕ_{cov} in 2/7, both concentrated among the methods most distant from PGSR (eRankGS, MiniSplatting). $\varepsilon_{\mathrm{res}}$ reaches significance in 1/7 (eRankGS only). This channel-selective pattern is consistent with the planar constraint acting precisely on opacity rather than producing a global optimisation shift, and demonstrates that the profiling framework captures design-intent granularity beyond the single channel tested in table 4.

Channel selectivity: designed structure vs. optimisation failure. If representational divergence were merely an optimisation artifact, all channels would shift together. We test this with paired Wilcoxon signed-rank tests for PGSR and 2DGS Huang et al. [2024] each vs. vanilla-3DGS on 12 Tanks & Temples scenes. Both methods separate unanimously on $\Delta\varepsilon_{\mathrm{opa}}$ (12/12, $p = 4.9 \times 10^{-4}$)

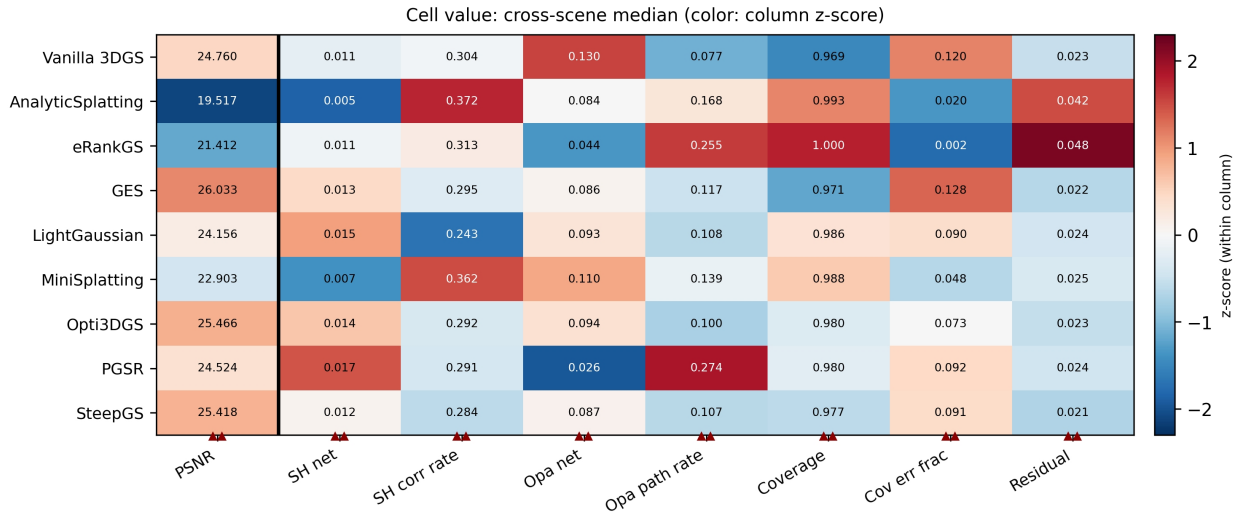


Figure 5: Counterfactual profile heatmap across methods. Columns include PSNR as an output-level reference, the six diagnostic channels, and an auxiliary raw coverage column used only to contextualise ϕ_{cov}. Cell values are cross-scene medians; colour encodes within-column z-score. The PSNR column is separated by a vertical rule as a reference baseline.

and ρ_{opa} (12/12)—a shared geometric signal. But the remaining fingerprints diverge: ϕ_{cov} shows 2DGS at 12/12 vs. PGSR at 9/12; $\Delta\varepsilon_{\text{SH}}$ directions are reversed between the two methods; ρ_{SH} and ε_{res} show opposite signs. Channel selectivity argues against a simple global-shift explanation: PGSR is 12/12 on $\Delta\varepsilon_{\text{opa}}$ but 8/12 on ρ_{SH} ($p = 0.27$); 2DGS is 12/12 on $\Delta\varepsilon_{\text{opa}}$ and ϕ_{cov} but 8/12 on $\Delta\varepsilon_{\text{SH}}$ ($p = 0.73$). Different geometric constraints produce overlapping but distinguishable channel fingerprints. *Caveat.* 2DGS uses surfel primitives; its profiling uses scale-padding (zeroing the third scale dimension; PSNR offset ≤ 0.3 dB on TRUCK, rank-order preserved). This is an approximation, not a strict equivalence; 2DGS results are supplementary evidence, not N_3-equivalent comparisons.

5.6 Paired Validation

Four paired comparisons against vanilla-3DGS test whether the channels track the directions stated in each variant's design (table 4; an additional pair, GES on ϕ_{cov}, is excluded from the primary set because GES's design does not explicitly target coverage; this pair is available in the released CSV).

PGSR on $\Delta\varepsilon_{\text{opa}}$ and eRankGS on ϕ_{cov} both reach unanimous direction agreement across all 31 scenes. eRankGS exhibits training failure across the entire NeRF-Synthetic dataset (8 scenes, cross-scene mean PSNR 17.68 dB; section 3.3 and Appendix D). A threshold sweep that progressively excludes scenes where eRankGS PSNR $< t$ yields direction agreement of 1.000 at every threshold tested (table 15), confirming that the unanimous result is not an artefact of counting broken reconstructions. Wilcoxon signed-rank tests against H_0: median $\Delta = 0$ confirm that all four medians are significant at the corpus level: P01 and P02 at the unanimous-direction floor of $p < 10^{-9}$, P04 at $p = 1.2 \times 10^{-6}$, and P03 at $p = 5.0 \times 10^{-7}$ in the direction *opposite* to LightGaussian's design intent.

LightGaussian Fan et al. [2024b] is the informative counter-finding: pruning, naturally expected to reduce opacity pathology, instead raises ρ_{opa} by a median +0.031 (3/31 scenes in the expected direction)—small in absolute terms but consistent in sign. We report this as a side effect the method's headline metrics did not flag, not as a failure of its design objective.

Ruling out the mechanical pruning account. A natural alternative explanation for the ρ_{opa} increase is that pruning mechanically forces surviving Gaussians toward non-binary opacity to maintain smooth blending. If this were the case, a higher pruning ratio (fewer surviving Gaussians)

Table 4: Paired validation against vanilla-3DGS. Direction agreement is the fraction of scenes whose delta sign matches the variant's stated design intent; p is the two-sided Wilcoxon signed-rank test against H_0: median $\Delta = 0$ on the 31 per-scene deltas. Full per-scene deltas in Appendix C (table 8). Summary statistics (Median Δ, IQR) are computed from full-precision values; the per-scene deltas in table 8 are rounded to three decimals, so re-deriving summaries from that table can yield ± 0.001 differences. †eRankGS exhibits method-level training failure on the NeRF-Synthetic dataset (cross-scene mean PSNR 17.68 dB; 8 scenes affected, not only LEGO). A threshold sweep excluding all scenes where eRankGS PSNR $< t$ preserves direction agreement at 1.000 for every tested threshold ($t \in \{17, 19, 21, 23, 25\}$ dB; n ranges from 31 down to 8; full table in table 15).

ID	Variant	Channel	Median Δ	IQR	Dir. agreement	p	Status
P01	PGSR	$\Delta\varepsilon_{\text{opa}}$	-0.095	0.143	1.000 (31/31)	$<10^{-9}$	Validated
P02	eRankGS	ϕ_{cov}	-0.118	0.268	1.000 (31/31)†	$<10^{-9}$	Validated
P03	LightGaussian	ρ_{opa}	$+0.031$	0.027	0.097 (3/31)	5.0×10^{-7}	Counter-finding
P04	SteepGS	$\Delta\varepsilon_{\text{opa}}$	-0.029	0.042	0.839 (26/31)	1.2×10^{-6}	Directional

should predict higher ρ_{opa}. We test this directly: across 31 scenes, the Spearman correlation between LightGaussian's retention ratio (LG count / vanilla count, ranging 19–38%) and $\Delta\rho_{\text{opa}}$ is $r = -0.040$ ($p = 0.828$). LightGaussian's absolute Gaussian count vs. ρ_{opa} is positively correlated ($r = +0.581$, $p = 0.0006$)—opposite to the pruning hypothesis—explained by scene-level confounding (larger scenes have more Gaussians and higher baseline pathology). Pruning ratio does not predict the pathology shift; the ρ_{opa} increase reflects a non-trivial opacity-distribution change not reducible to primitive count.

SteepGS Wang et al. [2025a] on $\Delta\varepsilon_{\text{opa}}$ reaches 0.839 (26/31), directionally consistent with steepness-aware densification reducing semi-transparent primitive contribution. Appendix G translates these signals into preliminary design-direction hypotheses.

SH channel validation. LightGaussian includes SH distillation (truncation of higher-order coefficients), providing a natural test case for the SH channels. Expected: the remaining low-order SH carries more per-pixel weight ($\Delta\varepsilon_{\text{SH}}$ increases) and over-fitting decreases (ρ_{SH} decreases). Observed: across 31 scenes, $\Delta\varepsilon_{\text{SH}}$ is positive in 26/31 and $\Delta\rho_{\text{SH}}$ is negative in 24/31, confirming clear majority directional agreement on both channels. Independently, 3DGS-DR Ye et al. [2024a]—an N_2 method that modifies the rendering pipeline to defer reflections—shows $\Delta\varepsilon_{\text{SH}}$ lower in 26/29 scenes and ρ_{SH} higher in 25/29. Two mechanistically distinct SH modifications both produce clear, consistent directional signals on the SH channels. The residual channel ε_{res} remains a catch-all with no direct design-intent pairing and is reported descriptively.

5.7 Reliability and Computational Cost

Four checks (table 5) verify rendering determinism, cross-rasteriser rank consistency (median Spearman $\rho = 1.000$), joint-vs-marginal additivity (interaction -0.7%), and SH-depth sensitivity. The full 279-cell sweep completes in under one GPU-hour on an NVIDIA RTX 4080 SUPER. Profiling can be attached post-hoc to any trained standard-primitive 3DGS model.

5.8 Limitations of profiling

Counterfactual profiling covers $N_3 = 9$ methods; cross-method comparisons further exclude AnalyticSplatting on rasteriser-backend grounds, leaving a comparable subset of 8 methods. Three channels have prespecified paired validation via table 4 ($\Delta\varepsilon_{\text{opa}}$, ρ_{opa}, ϕ_{cov}). Two additional channels ($\Delta\varepsilon_{\text{SH}}$, ρ_{SH}) have post-hoc directional evidence from the SH distillation and deferred-reflection analyses above, but lack table 4-style prespecified paired validation. ε_{res} remains a descriptive

Table 5: Reliability checks. L_1 noise floor, cross-rasteriser rank consistency, joint-vs-marginal additivity, and SH-depth sensitivity.

Row	Check	Key result	Verdict
1	Rendering noise floor	L_1 $p_{99} = 0$; PSNR-equivalent ≥ 160 dB	deterministic
2	Rasteriser rank consistency	median $\rho = 1.000$ ($n = 19$; min 0.700)	consistent
3	Joint vs. marginal additivity	interaction -0.7% of marginal sum	approx. additive
4	SH-depth sensitivity	$\ell = 1$ contributes 17%; direction preserved	direction-consistent

catch-all with no empirical support. The additivity assumption is approximate (-0.7% interaction), so single-channel deltas carry small unresolved coupling terms.

6 Discussion and Conclusion

The empirical findings of section 4 together suggest that representational divergence between methods is real and method-specific, but does not reduce to the low-order organising principles tested. Three observations follow from the data, framed as tentative rather than normative: the simplest cell-level SH–loss trade-off signature is not observed—SH energy and training loss co-vary positively across scenes (with training loss as a difficulty proxy) rather than trading off (appendix F), though finer-grained (per-pixel or geometry-conditioned) signatures remain untested; several counterfactual channels exhibit greater method-level dispersion than PSNR even in the MSE domain within the saturated cluster (up to 3.7× for opacity pathology rate; table 3); and standard family boundaries do not correspond to the axes along which methods diverge dynamically (ARI = 0.025; section 4.4).

PSNR, SSIM, and LPIPS remain serviceable output-level references but carry limited discriminative power within the saturated cluster (section 4.1). Counterfactual profiling (section 5) supplies a complementary representational dimension rather than a replacement: within the saturated cluster, channels exhibit relative IQR up to 22.8× that of PSNR in the dB domain (3.7× in the MSE domain), and two of four paired comparisons against vanilla reach direction agreement of 1.000, with a third at 0.839 (section 5.6). The specific operators defined here are tied to standard 3DGS primitives; operator harmonisation across non-standard primitives (anchor, surfel, anti-aliased rasterisation) remains an open methodological problem, and we do not prescribe broader adoption ahead of community consensus on operator definitions. In the near term, counterfactual profiling sits alongside existing photometric reporting as a supplementary diagnostic, attached to whatever a paper already reports rather than displacing it.

Downstream validation: surface reconstruction. To test whether counterfactual channels predict task-relevant quality beyond PSNR, we extract meshes for four methods (vanilla-3DGS, 2DGS, GOF, PGSR; 2DGS and GOF are N_2 methods excluded from the main N_3 profiling on primitive-divergence grounds; for this evaluation the opacity-clamp operator is applied to their standard 3DGS opacity parameters—2DGS via scale-padding as in section 5.5, GOF via its unmodified opacity parameter) using each method's own mesh-extraction code on three Tanks & Temples scenes with ground-truth meshes (BARN, CATERPILLAR, TRUCK) and compute F1 scores using the official Tanks & Temples evaluation toolbox Knapitsch et al. [2017]. $\Delta\varepsilon_{\text{opa}}$ correlates with F1 (Spearman $r = -0.764$, $p = 0.004$, $n = 12$ cells in a 4-method × 3-scene crossed design); ρ_{opa} likewise ($r = +0.680$, $p = 0.015$). PSNR shows no correlation ($r = +0.063$, $p = 0.85$). The ranking divergence is concrete: vanilla-3DGS has the highest PSNR (25.05 dB) but the lowest F1 (13.3); PGSR's PSNR (24.54 dB) exceeds 2DGS (23.83 dB), yet F1 ranks PGSR (50.7) ≈ GOF (50.3) above 2DGS (36.7). The $\Delta\varepsilon_{\text{opa}}$ ordering (3DGS > 2DGS > GOF ≈ PGSR) aligns with F1; PSNR ordering does not. For users who care about geometry quality, PSNR shows no association with F1 in this pilot evaluation ($n = 12$ cells); $\Delta\varepsilon_{\text{opa}}$ shows a clear cell-level association ($r = -0.764$, $p = 0.004$; the crossed design

limits effective degrees of freedom). This is one instance of the recommended usage workflow: task (surface reconstruction) → channel ($\Delta\varepsilon_{\text{opa}}$) → method ranking that diverges from PSNR and aligns with task-relevant quality. Additional task instances (editing, compression) are explicit future work.

Why task-independent aggregation fails. A single aggregate score across counterfactual channels has no task-independent default: the "good" direction of the same channel reverses across tasks. Low $\Delta\varepsilon_{\text{opa}}$ is a positive signal for surface reconstruction (opacity tends toward binary, geometry hugs the surface) but a negative signal for translucent-scene rendering (semi-transparent primitives carry real information). High coverage is desirable for novel-view synthesis but may cause hallucination in 3D editing. This is analogous to precision and recall, for which no task-independent aggregation is universally optimal—even F1 encodes a specific trade-off between the two.

Recommended evaluation workflow. We recommend *task-conditioned channel selection*: identify the downstream task, select the relevant channel subset, and compare methods on that subset. The surface-reconstruction chain above is a complete instance: task → $\Delta\varepsilon_{\text{opa}} \rightarrow r = -0.764$ with F1, while PSNR shows no association. Table 14 provides preliminary channel–task mappings: geometric constraint → $\Delta\varepsilon_{\text{opa}}$; coverage control → ϕ_{cov}; compression quality audit → ρ_{opa} (LightGaussian's counter-finding as a warning). In practice, a paper proposing a new 3DGS method would report PSNR/SSIM/LPIPS alongside a six-channel counterfactual profile (the profiling tool attaches post-hoc to any trained standard-primitive 3DGS model in under one GPU-hour; section 5.7). Readers select channels according to their task. Currently only the surface-reconstruction mapping has experimental validation; establishing additional mappings is explicit future work.

Limitations. SplatAtlas covers static reconstruction only; findings are not claimed to extrapolate to 4D or dynamic 3DGS settings. All methods are trained under photometric supervision; statements about limited representational signal under photometric supervision do not extend to depth-, normal-, or geometry-prior-supervised variants. The benchmark is restricted to reconstruction from posed images: generative 3DGS, score-distillation pipelines, and text-to-3D methods are out of scope. Scene scale is capped at Mip-NeRF 360 outdoor extents; no city-scale, aerial, or SLAM-style reconstructions are included, and findings about saturation and representational divergence should not be assumed to transfer to settings where scene scale itself drives method behaviour. All quantitative results are conditional on the 31 scenes in section 3.2; extrapolation to transparent, highly reflective, weak-texture, or extreme-lighting regimes is not asserted. Counterfactual profiling covers $N_3 = 9$ methods $\times$ 31 scenes; extending to the broader method inventory beyond N_3 would require substantially more computation, with a rough extrapolation from the current N_3 profiling sweep giving approximately 3.4$\times$ the cell count (961/279); the actual cost multiplier may be higher if operator definitions require per-method adaptation for non-standard primitives. The counterfactual deltas in section 5 are post-hoc descriptive interventions on a fixed rendering model and do not constitute causal identification; they are representational diagnostics, and no mechanistic claims follow from their magnitudes. All counterfactual comparisons are conducted within a single backend (gsplat); cross-method conclusions depend on within-backend rank consistency, not on absolute agreement between gsplat and native rasterisers. Methods with non-standard primitives (anchor-based, surfel-based, anti-aliased rasterisation) carry comparability caveats documented in sections 3.2 and 5.8; broader operator harmonisation across primitive variants, prespecified paired validation for $\Delta\varepsilon_{\text{SH}}$ and ρ_{SH} (which currently have only post-hoc directional evidence), and any validation for ε_{res} remain future work. Method-level training anomalies under the unified default-hyperparameter protocol are documented in appendix E.1 and affect approximately 8% of cells. The ANOVA variance shares in section 4.2 include approximately 8% of cells with method-level training anomalies; the method-attributed fractions for opacity and anisotropy may be inflated by these cells; formal exclusion-sensitivity analysis is deferred to future work.

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A Method Inventory

This appendix lists the 31 methods evaluated in SplatAtlas, grouped by category and ordered alphabetically by display name within each category. The Tier column records the highest analytic tier (section 3.2) at which each method participates: methods at N_2 contribute to the trajectory and end-state analyses (section 4) but are excluded from counterfactual profiling on primitive-divergence grounds; methods at N_3 enter the counterfactual analyses of section 5; the comparable subset of N_3 used for cross-method counterfactual comparisons in sections 5.4 and 5.5 excludes `analyticsplatting` on rasteriser-backend grounds, leaving 8 methods.

Category labels in table 6 are design-intent annotations assigned by the authors based on each method's primary stated motivation; they are bookkeeping rather than data-driven cluster assignments, and section 4.4 reports that data-driven trajectory clusters are nearly orthogonal to this five-category taxonomy (ARI = 0.025, NMI = 0.061). Quantitative claims in this paper do not rest on these labels.

Note explanations.

baseline Reference method against which paired counterfactual comparisons in section 5.6 are computed.

rasteriser caveat `analyticsplatting` uses a custom anti-aliased rasteriser. It enters N_3 but is excluded from cross-method counterfactual comparisons on the rasteriser-backend grounds documented in sections 3.2 and 5.8; its per-cell seed variance on LEGO is also flagged (Appendix D).

NeRF-Synthetic failure `erankgs` exhibits method-level training failure across the NeRF-Synthetic dataset (cross-scene mean PSNR 17.68 dB; section 3.3, Appendix E.1). The LEGO cell is excluded from seed-variance statistics; the method is retained in benchmark tables as a failure-mode exemplar.

B Scene Inventory

This appendix lists the 31 scenes used in SplatAtlas, grouped by dataset. Two split conventions apply. For NeRF-Synthetic scenes, train, test, and validation counts are read directly from the verified local `transforms_train.json`, `transforms_test.json`, and `transforms_val.json` files used in this release; all eight scenes resolve to the same local split sizes (320/80/80). This local split is reported explicitly here and should not be read as a claim that SplatAtlas uses the original NeRF-Synthetic train/test protocol. For COLMAP Schönberger and Frahm [2016]-style real-world scenes (Tanks & Temples, Mip-NeRF 360, Deep Blending), train and test counts follow the project's hold-every-eighth-frame evaluation split: every 8th frame is held out as test, the remainder as train, matching the standard 3DGS `llffhold` = 8 convention. A validation split is not defined for these scenes, entered as "—".

The Resolution column reports the per-axis downsampling factor used during training and evaluation: 1 for synthetic scenes and Deep Blending, 2 for indoor Mip-NeRF 360 and Tanks & Temples, and 4 for outdoor Mip-NeRF 360.

Table 6: Method inventory of SplatAtlas. Tier indicates the highest analytic tier the method participates in (section 3.2); the comparable subset of N_3 used for cross-method counterfactual comparisons further excludes `analyticsplatting` on rasteriser-backend grounds. Notes flag method-level caveats that recur in the main text. Reference column gives the canonical publication for each method; bibliographic details in the references.

Method (key)	Display name	Reference	Category	Tier	Notes
FREE					
`3dgsmcmc`	3DGS-MCMC	[Kheradmand et al., 2024]	FREE	N_2	
`absgs`	AbsGS	[Ye et al., 2024b]	FREE	N_2	
`atomgs`	AtomGS	[Liu et al., 2024]	FREE	N_2	
`conegs`	ConeGS	[Baranowski et al., 2025]	FREE	N_2	
`ges`	GES	[Hamdi et al., 2024]	FREE	N_3	
`ghap`	GHAP	[Wang et al., 2025b]	FREE	N_2	
`gslpm`	GSLPM	[Yang et al., 2025a]	FREE	N_2	
`lapisgs`	LapisGS	[Shi et al., 2025]	FREE	N_2	
`reactgs`	ReactGS	[Cheng et al., 2025]	FREE	N_2	
`vanilla_3dgs`	3DGS	[Kerbl et al., 2023]	FREE	N_3	baseline
GEO_REG					
`2dgs`	2DGS	[Huang et al., 2024]	GEO_REG	N_2	
`gaussian_surfel`	Gaussian Surfels	[Dai et al., 2024]	GEO_REG	N_2	
`gof`	GOF	[Yu et al., 2024b]	GEO_REG	N_2	
`pgsr`	PGSR	[Chen et al., 2025a]	GEO_REG	N_3	
`scaffoldgs`	Scaffold-GS	[Lu et al., 2024]	GEO_REG	N_2	
PRIM_CTRL					
`coadaptgs`	CoAdaptGS	[Chen et al., 2025b]	PRIM_CTRL	N_2	
`erankgs`	eRankGS	[Hyung et al., 2024]	PRIM_CTRL	N_3	NeRF-Synthetic failure
`gaussianpro`	GaussianPro	[Cheng et al., 2024]	PRIM_CTRL	N_2	
`minisplatting`	MiniSplatting	[Fang and Wang, 2024]	PRIM_CTRL	N_3	
`opti3dgs`	Opti3DGS	[Farooq et al., 2025]	PRIM_CTRL	N_3	
`steepgs`	SteepGS	[Wang et al., 2025a]	PRIM_CTRL	N_3	
RENDER					
`3dgs_dr`	3DGS-DR	[Ye et al., 2024a]	RENDER	N_2	
`analyticsplatting`	AnalyticSplatting	[Liang et al., 2024]	RENDER	N_3	rasteriser caveat
`lod_gs`	LoD-GS	[Yang et al., 2025b]	RENDER	N_2	
`mipsplatting`	Mip-Splatting	[Yu et al., 2024a]	RENDER	N_2	
`pixelgs`	PixelGS	[Zhang et al., 2024]	RENDER	N_2	
COMPRESS					
`cdcgs`	CDCGS	[Dong et al., 2025]	COMPRESS	N_2	
`hogs`	HoGS	[Liu et al., 2025]	COMPRESS	N_2	
`lightgaussian`	LightGaussian	[Fan et al., 2024b]	COMPRESS	N_3	
`octree_gs`	Octree-GS	[Ren et al., 2024]	COMPRESS	N_2	
`trimgs`	TrimGS	[Fan et al., 2024a]	COMPRESS	N_2	

C Paired Counterfactual Validation: Full Table

This appendix provides the per-scene paired counterfactual deltas for the four paired comparisons summarised in table 4, totalling 124 rows (4 pairs × 31 scenes). Each row corresponds to a (`pair_id`, variant, scene) tuple and carries the per-pixel deltas evaluated as variant minus base, with base fixed to `vanilla_3dgs` for all four pairs. Per-scene deltas are restricted to rows with `pair_id` ∈ {P01, P02, P03, P04}. An auxiliary fifth pair, P05: `ges` on $\Delta\phi_{\text{cov}}$, is available in the released CSV but

Table 7: Scene inventory of SplatAtlas, grouped by dataset. Train/test/validation counts are read directly from local split files for NeRF-Synthetic and computed via the hold-every-eighth-frame convention for the COLMAP-style real-world datasets. Resolution reports the per-axis downsampling factor used during training and evaluation.

Scene	Dataset	#Train	#Test	#Val	#Total	Resolution	Type
Tanks & Temples							
AUDITORIUM	T&T	260	38	—	298	2	real, indoor
BALLROOM	T&T	283	41	—	324	2	real, indoor
BARN	T&T	358	52	—	410	2	real, outdoor
CATERPILLAR	T&T	335	48	—	383	2	real, outdoor
COURTROOM	T&T	263	38	—	301	2	real, indoor
LIGHTHOUSE	T&T	270	39	—	309	2	real, outdoor
MUSEUM	T&T	263	38	—	301	2	real, indoor
PALACE	T&T	438	63	—	501	2	real, indoor
PLAYGROUND	T&T	268	39	—	307	2	real, outdoor
TEMPLE	T&T	264	38	—	302	2	real, indoor
TRAIN	T&T	263	38	—	301	2	real, outdoor
TRUCK	T&T	219	32	—	251	2	real, outdoor
Mip-NeRF 360							
BICYCLE	Mip-NeRF 360	169	25	—	194	4	real, outdoor
BONSAI	Mip-NeRF 360	255	37	—	292	2	real, indoor
COUNTER	Mip-NeRF 360	210	30	—	240	2	real, indoor
FLOWERS	Mip-NeRF 360	151	22	—	173	4	real, outdoor
GARDEN	Mip-NeRF 360	161	24	—	185	4	real, outdoor
KITCHEN	Mip-NeRF 360	244	35	—	279	2	real, indoor
ROOM	Mip-NeRF 360	272	39	—	311	2	real, indoor
STUMP	Mip-NeRF 360	109	16	—	125	4	real, outdoor
TREEHILL	Mip-NeRF 360	123	18	—	141	4	real, outdoor
NeRF-Synthetic							
CHAIR	NeRF-Synthetic	320	80	80	480	1	synthetic, object
DRUMS	NeRF-Synthetic	320	80	80	480	1	synthetic, object
FICUS	NeRF-Synthetic	320	80	80	480	1	synthetic, object
HOTDOG	NeRF-Synthetic	320	80	80	480	1	synthetic, object
LEGO	NeRF-Synthetic	320	80	80	480	1	synthetic, object
MATERIALS	NeRF-Synthetic	320	80	80	480	1	synthetic, object
MIC	NeRF-Synthetic	320	80	80	480	1	synthetic, object
SHIP	NeRF-Synthetic	320	80	80	480	1	synthetic, object
Deep Blending							
DRJOHNSON	Deep Blending	231	33	—	264	1	real, indoor
PLAYROOM	Deep Blending	197	29	—	226	1	real, indoor

is excluded from the primary validation set because GES's stated design motivation (generalised exponential primitives) does not explicitly target scene coverage; we do not tabulate it here.

The direction-agreement convention used by table 4 fixes the expected sign per pair, for example $\Delta\varepsilon_{\text{opa}}$ expected negative for `pgsr`, and $\Delta\rho_{\text{opa}}$ expected negative under `lightgaussian`'s pruning intent, with the observed reverse direction logged as a counter-finding in the main text. Individual extreme deltas in the per-scene breakdown may reflect scene-specific characteristics rather than systematic properties of the variant: notably, FICUS produces an unusually high ϕ_{cov} baseline (0.74 on `vanilla_3dgs`; section 5.2) that propagates into several rows below. The `erankgs` × LEGO

cell carries the METHOD_FAILURE annotation from section 3.3 and should be read with the corresponding caveat.

Backend caveat. All counterfactual deltas in table 8 are computed under the unified `gsplat` backend (section 5.2). ΔPSNR values in this table are therefore backend-consistent across methods but are not directly comparable in absolute magnitude to the native-rasteriser PSNRs of tables 9 and 11: native rasterisers may render the same trained Gaussian field at meaningfully different absolute PSNR, and on individual cells (notably LEGO) the gap can reach several dB. Backend rank-order consistency is verified in table 5 (median Spearman $\rho = 1.000$ across 19 scenes); the absolute-magnitude offset is the price of using a single rendering backend so that channel-level interventions remain comparable across methods.

Table 8: Per-scene paired counterfactual deltas for the four pairs in table 4. Base method is `vanilla_3dgs` for all rows. Channel-delta columns: ΔPSNR, $\Delta\varepsilon_{\mathrm{SH}}$, $\Delta\rho_{\mathrm{SH}}$, $\Delta\varepsilon_{\mathrm{opa}}$, $\Delta\rho_{\mathrm{opa}}$, $\Delta\phi_{\mathrm{cov}}$, $\Delta\varepsilon_{\mathrm{res}}$. All deltas reported to 3-decimal precision (P01–P04 subset, 124 rows). All deltas computed under the unified `gsplat` backend; see the "Backend caveat" paragraph above.

Pair	Variant	Scene	ΔPSNR	$\Delta\varepsilon_{\mathrm{SH}}$	$\Delta\rho_{\mathrm{SH}}$	$\Delta\varepsilon_{\mathrm{opa}}$	$\Delta\rho_{\mathrm{opa}}$	$\Delta\phi_{\mathrm{cov}}$	$\Delta\varepsilon_{\mathrm{res}}$
P01	PGSR	Auditorium	-0.167	+0.008	-0.040	-0.123	+0.244	-0.003	-0.003
P01	PGSR	Ballroom	-0.640	+0.007	-0.019	-0.176	+0.149	+0.007	-0.000
P01	PGSR	Barn	-0.934	+0.001	+0.043	-0.172	+0.286	-0.554	-0.001
P01	PGSR	Bicycle	-0.074	+0.008	-0.057	-0.141	+0.166	-0.005	+0.001
P01	PGSR	Bonsai	-0.721	+0.006	-0.018	-0.095	+0.179	-0.276	-0.000
P01	PGSR	Caterpillar	-0.408	+0.015	-0.020	-0.207	+0.247	-0.029	-0.009
P01	PGSR	Chair	-0.900	+0.001	+0.003	-0.005	+0.017	+0.012	+0.003
P01	PGSR	Counter	-1.081	+0.008	-0.020	-0.043	+0.167	-0.150	+0.001
P01	PGSR	Courtroom	+0.128	+0.008	-0.030	-0.209	+0.197	+0.005	-0.008
P01	PGSR	DrJohnson	-1.112	+0.002	-0.004	-0.087	+0.261	-0.036	+0.001
P01	PGSR	Drums	-0.250	+0.001	-0.003	-0.013	+0.026	-0.016	+0.007
P01	PGSR	Ficus	-0.986	+0.001	+0.001	-0.005	+0.014	-0.017	+0.010
P01	PGSR	Flowers	-0.178	+0.002	-0.001	-0.089	+0.114	-0.015	-0.002
P01	PGSR	Garden	-0.444	+0.009	-0.026	-0.045	+0.035	+0.121	+0.001
P01	PGSR	Hotdog	+2.874	+0.005	-0.006	-0.002	+0.040	-0.156	+0.003
P01	PGSR	Kitchen	-1.087	+0.009	-0.018	-0.035	+0.092	+0.122	+0.001
P01	PGSR	Lego	-0.580	+0.001	+0.003	-0.005	+0.029	-0.042	+0.005
P01	PGSR	Lighthouse	-0.539	+0.011	+0.033	-0.206	+0.305	-0.020	-0.004
P01	PGSR	Materials	-1.169	+0.002	+0.001	-0.006	+0.015	+0.039	+0.002
P01	PGSR	Mic	+0.555	+0.001	+0.002	-0.002	+0.006	-0.011	+0.002
P01	PGSR	Museum	-0.105	+0.008	-0.038	-0.135	+0.195	-0.004	-0.005
P01	PGSR	Palace	-2.387	+0.005	+0.010	-0.161	+0.249	-0.007	+0.025
P01	PGSR	Playground	-0.928	+0.003	-0.011	-0.201	+0.260	+0.018	-0.002
P01	PGSR	Playroom	+0.055	+0.003	-0.045	-0.110	+0.254	+0.010	-0.002
P01	PGSR	Room	-1.375	+0.004	-0.016	-0.034	+0.163	+0.025	+0.001
P01	PGSR	Ship	+1.936	+0.003	-0.024	-0.008	+0.009	-0.366	+0.002
P01	PGSR	Stump	-0.015	+0.003	-0.019	-0.046	+0.079	-0.042	-0.002

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Pair	Variant	Scene	ΔPSNR	$\Delta\varepsilon_{\mathrm{SH}}$	$\Delta\rho_{\mathrm{SH}}$	$\Delta\varepsilon_{\mathrm{opa}}$	$\Delta\rho_{\mathrm{opa}}$	$\Delta\phi_{\mathrm{cov}}$	$\Delta\varepsilon_{\mathrm{res}}$
P01	PGSR	Temple	-1.935	+0.004	+0.008	-0.249	+0.304	-0.079	+0.010
P01	PGSR	Train	-0.844	+0.014	-0.028	-0.207	+0.256	-0.028	+0.001
P01	PGSR	Treehill	-0.410	+0.005	-0.017	-0.115	+0.164	-0.100	+0.001
P01	PGSR	Truck	-0.375	+0.005	-0.013	-0.131	+0.207	-0.120	+0.002
P02	eRankGS	Auditorium	-1.251	+0.004	-0.048	-0.113	+0.182	-0.017	+0.017
P02	eRankGS	Ballroom	-1.519	+0.003	-0.011	-0.203	+0.171	-0.070	-0.002
P02	eRankGS	Barn	-1.211	+0.000	+0.046	-0.124	+0.143	-0.690	+0.003
P02	eRankGS	Bicycle	-1.147	+0.002	-0.008	-0.120	+0.141	-0.442	+0.002
P02	eRankGS	Bonsai	-0.321	-0.002	+0.005	-0.098	+0.159	-0.426	-0.001
P02	eRankGS	Caterpillar	-1.432	+0.007	-0.009	-0.145	+0.132	-0.118	-0.003
P02	eRankGS	Chair	-32.934	-0.003	+0.166	+0.057	+0.275	-0.026	+0.171
P02	eRankGS	Counter	-0.718	+0.001	-0.002	-0.046	+0.136	-0.210	+0.000
P02	eRankGS	Courtroom	-1.995	+0.002	+0.002	-0.196	+0.199	-0.020	+0.006
P02	eRankGS	DrJohnson	-1.907	+0.001	+0.005	-0.077	+0.193	-0.103	+0.004
P02	eRankGS	Drums	-22.222	-0.003	+0.026	+0.024	+0.205	-0.055	+0.138
P02	eRankGS	Ficus	-21.717	-0.003	-0.005	+0.016	+0.227	-0.022	+0.260
P02	eRankGS	Flowers	-0.464	-0.001	+0.010	-0.088	+0.121	-0.162	-0.001
P02	eRankGS	Garden	-0.593	+0.001	-0.007	-0.060	+0.107	-0.442	+0.002
P02	eRankGS	Hotdog	-25.758	+0.003	+0.171	+0.054	+0.152	-0.443	+0.195
P02	eRankGS	Kitchen	-2.711	+0.002	+0.013	-0.041	+0.145	-0.261	+0.012
P02	eRankGS	Lego	-25.052	-0.000	+0.047	+0.052	+0.231	-0.362	+0.158
P02	eRankGS	Lighthouse	-5.405	-0.008	+0.090	-0.151	+0.234	-0.033	+0.029
P02	eRankGS	Materials	-25.281	-0.009	+0.051	+0.023	+0.204	-0.215	+0.102
P02	eRankGS	Mic	-25.184	-0.004	+0.117	+0.011	+0.316	-0.051	+0.175
P02	eRankGS	Museum	-0.821	+0.001	-0.014	-0.108	+0.143	-0.027	-0.001
P02	eRankGS	Palace	-6.109	-0.009	+0.077	-0.153	+0.272	-0.018	+0.129
P02	eRankGS	Playground	-1.202	+0.002	-0.010	-0.116	+0.129	-0.102	-0.003
P02	eRankGS	Playroom	-1.349	+0.003	-0.026	-0.114	+0.252	-0.075	+0.001
P02	eRankGS	Room	-4.251	+0.002	-0.012	-0.026	+0.102	-0.159	+0.040
P02	eRankGS	Ship	-20.504	+0.002	+0.167	-0.010	+0.088	-0.586	+0.305
P02	eRankGS	Stump	-0.649	+0.000	+0.004	-0.027	+0.086	-0.415	-0.001
P02	eRankGS	Temple	-6.685	-0.004	+0.057	-0.201	+0.221	-0.114	+0.091
P02	eRankGS	Train	-2.130	+0.000	+0.013	-0.149	+0.173	-0.094	+0.004
P02	eRankGS	Treehill	-1.124	-0.001	+0.025	-0.120	+0.181	-0.280	+0.004
P02	eRankGS	Truck	-1.936	+0.003	-0.024	-0.144	+0.208	-0.257	+0.002
P03	LightGaussian	Auditorium	-2.421	-0.004	+0.002	-0.047	+0.059	-0.016	+0.019
P03	LightGaussian	Ballroom	-1.681	+0.010	-0.063	-0.097	+0.063	-0.060	+0.010
P03	LightGaussian	Barn	-1.192	+0.009	-0.022	-0.080	+0.047	-0.634	+0.000
P03	LightGaussian	Bicycle	-0.134	+0.004	-0.043	-0.053	+0.050	-0.146	-0.001
P03	LightGaussian	Bonsai	-0.869	+0.009	-0.059	-0.031	+0.043	-0.048	+0.002
P03	LightGaussian	Caterpillar	-0.756	+0.004	-0.039	-0.045	+0.018	-0.049	+0.005
P03	LightGaussian	Chair	+1.255	+0.000	+0.006	-0.000	+0.007	+0.045	+0.001
P03	LightGaussian	Counter	-0.681	+0.006	-0.050	-0.021	+0.040	-0.100	+0.001
P03	LightGaussian	Courtroom	-2.401	+0.007	-0.061	-0.091	+0.054	-0.017	+0.023

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Pair	Variant	Scene	ΔPSNR	$\Delta\varepsilon_{\text{SH}}$	$\Delta\rho_{\text{SH}}$	$\Delta\varepsilon_{\text{opa}}$	$\Delta\rho_{\text{opa}}$	$\Delta\phi_{\text{cov}}$	$\Delta\varepsilon_{\text{res}}$
P03	LightGaussian	DrJohnson	-2.357	+0.008	-0.098	-0.026	+0.071	-0.096	+0.015
P03	LightGaussian	Drums	+0.839	-0.001	+0.028	+0.008	+0.008	+0.083	-0.002
P03	LightGaussian	Ficus	+8.912	-0.001	-0.002	+0.009	-0.011	+0.707	-0.023
P03	LightGaussian	Flowers	-0.207	+0.006	-0.064	-0.021	+0.032	-0.031	+0.001
P03	LightGaussian	Garden	-0.596	+0.003	-0.041	+0.014	-0.010	+0.065	+0.002
P03	LightGaussian	Hotdog	+0.024	+0.001	+0.015	-0.001	+0.008	-0.354	+0.000
P03	LightGaussian	Kitchen	-0.612	+0.003	-0.022	-0.011	+0.025	-0.151	+0.002
P03	LightGaussian	Lego	-3.360	+0.001	+0.012	-0.004	+0.027	-0.202	+0.001
P03	LightGaussian	Lighthouse	-2.700	+0.001	-0.002	-0.083	+0.020	-0.027	+0.034
P03	LightGaussian	Materials	+1.370	-0.000	+0.006	-0.008	+0.029	+0.171	+0.001
P03	LightGaussian	Mic	+1.781	-0.000	+0.031	+0.012	+0.003	+0.115	+0.000
P03	LightGaussian	Museum	-0.606	+0.006	-0.055	-0.046	+0.046	-0.011	+0.004
P03	LightGaussian	Palace	-3.451	+0.001	-0.056	-0.092	+0.130	-0.015	+0.060
P03	LightGaussian	Playground	-1.297	+0.008	-0.053	-0.043	+0.023	-0.061	+0.005
P03	LightGaussian	Playroom	-2.534	+0.008	-0.146	-0.030	+0.032	-0.051	+0.007
P03	LightGaussian	Room	-1.549	+0.015	-0.145	-0.014	+0.029	-0.103	+0.008
P03	LightGaussian	Ship	+3.585	+0.001	-0.003	-0.004	-0.027	-0.307	-0.001
P03	LightGaussian	Stump	+0.034	+0.004	-0.033	-0.036	+0.044	-0.111	-0.001
P03	LightGaussian	Temple	-1.767	+0.006	-0.028	-0.094	+0.044	-0.089	+0.015
P03	LightGaussian	Train	-0.923	+0.004	-0.044	-0.051	+0.027	-0.040	+0.008
P03	LightGaussian	Treehill	-0.154	+0.008	-0.063	-0.069	+0.076	-0.159	+0.007
P03	LightGaussian	Truck	-0.344	+0.008	-0.061	-0.056	+0.045	-0.125	+0.000
P04	SteepGS	Auditorium	-0.379	+0.000	-0.001	-0.040	+0.046	-0.011	+0.002
P04	SteepGS	Ballroom	-0.280	+0.003	-0.008	-0.084	+0.045	-0.047	-0.001
P04	SteepGS	Barn	-0.327	+0.001	+0.024	-0.069	+0.045	-0.635	-0.003
P04	SteepGS	Bicycle	-0.107	+0.002	-0.016	-0.013	+0.023	-0.271	-0.000
P04	SteepGS	Bonsai	+0.232	-0.001	-0.004	-0.029	+0.031	-0.280	-0.001
P04	SteepGS	Caterpillar	-0.159	+0.008	-0.028	-0.044	+0.028	-0.045	-0.001
P04	SteepGS	Chair	+3.196	+0.000	+0.004	-0.002	+0.009	+0.068	-0.001
P04	SteepGS	Counter	-0.085	+0.000	-0.002	-0.024	+0.053	-0.139	-0.000
P04	SteepGS	Courtroom	-0.356	+0.002	-0.009	-0.056	+0.030	-0.014	+0.002
P04	SteepGS	DrJohnson	-0.422	+0.001	+0.005	-0.029	+0.057	-0.080	+0.001
P04	SteepGS	Drums	+2.657	-0.000	-0.005	-0.004	+0.002	+0.128	-0.003
P04	SteepGS	Ficus	+8.211	-0.001	-0.005	+0.012	-0.014	+0.644	-0.023
P04	SteepGS	Flowers	-0.155	+0.002	-0.009	+0.011	+0.027	-0.052	-0.000
P04	SteepGS	Garden	-0.099	+0.001	-0.006	-0.036	+0.034	-0.274	+0.000
P04	SteepGS	Hotdog	+4.250	+0.001	-0.004	+0.000	-0.006	-0.053	-0.001
P04	SteepGS	Kitchen	-0.155	+0.002	-0.006	-0.012	+0.023	-0.175	+0.001
P04	SteepGS	Lego	-3.974	+0.000	-0.004	+0.000	+0.008	-0.170	-0.001
P04	SteepGS	Lighthouse	-0.612	+0.006	+0.010	-0.039	+0.019	-0.018	+0.011
P04	SteepGS	Materials	+3.533	+0.001	-0.007	-0.009	+0.013	+0.211	-0.000
P04	SteepGS	Mic	+8.531	+0.000	-0.005	-0.001	-0.002	+0.395	-0.003
P04	SteepGS	Museum	+0.143	+0.002	-0.012	-0.029	+0.031	-0.018	-0.003
P04	SteepGS	Palace	-1.242	+0.001	+0.003	-0.046	+0.034	-0.012	+0.022

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Pair	Variant	Scene	ΔPSNR	$\Delta\varepsilon_{\mathrm{SH}}$	$\Delta\rho_{\mathrm{SH}}$	$\Delta\varepsilon_{\mathrm{opa}}$	$\Delta\rho_{\mathrm{opa}}$	$\Delta\phi_{\mathrm{cov}}$	$\Delta\varepsilon_{\mathrm{res}}$
P04	SteepGS	Playground	-0.035	+0.004	-0.020	-0.037	+0.020	-0.059	-0.003
P04	SteepGS	Playroom	-0.283	+0.000	-0.003	-0.059	+0.072	-0.060	-0.000
P04	SteepGS	Room	-0.101	+0.001	-0.014	-0.014	+0.047	-0.080	-0.000
P04	SteepGS	Ship	+5.190	+0.001	-0.017	+0.007	-0.034	-0.086	-0.002
P04	SteepGS	Stump	-0.301	+0.001	-0.002	-0.028	+0.046	-0.267	-0.001
P04	SteepGS	Temple	-0.413	+0.002	+0.008	-0.091	+0.051	-0.081	-0.003
P04	SteepGS	Train	-0.037	+0.007	-0.045	-0.050	+0.034	-0.029	-0.001
P04	SteepGS	Treehill	-0.195	+0.002	-0.001	-0.075	+0.091	-0.170	+0.001
P04	SteepGS	Truck	-0.122	+0.002	-0.023	-0.057	+0.050	-0.141	-0.001

D Seed Variance: Full Data

This appendix provides the full data underlying the seed-variance protocol of section 3.3. Coverage: two scenes (bonsai, lego) crossed with the 10 methods of N_2 for which seed runs completed, three seeds per (method, scene) cell, comprising 60 raw rows. Table 9 reports the per-cell summary; Table 10 reports the Kendall τ values.

Table 9: Per-(method, scene) seed-variance summary; $n_{\mathrm{seeds}} = 3$ for all cells. Outlier flag marks cells with σ exceeding the cohort P_{95} (0.384 dB, including the analyticsplatting × lego cell). `erankgs` × lego carries METHOD_FAILURE status and is excluded from cohort statistics.

Method	Scene	$\overline{\mathrm{PSNR}}$	σ_{PSNR}	min	max	Status
`3dgsmcmc`	Bonsai	32.763	0.084	32.677	32.845	—
`analyticsplatting`	Bonsai	32.162	0.279	31.999	32.484	—
`erankgs`	Bonsai	31.762	0.035	31.722	31.787	—
`ges`	Bonsai	31.806	0.064	31.733	31.855	—
`lightgaussian`	Bonsai	31.495	0.038	31.461	31.537	—
`minisplatting`	Bonsai	31.587	0.005	31.582	31.592	—
`opti3dgs`	Bonsai	32.342	0.058	32.280	32.395	—
`pgsr`	Bonsai	31.347	0.016	31.338	31.366	—
`steepgs`	Bonsai	32.330	0.038	32.288	32.361	—
`vanilla_3dgs`	Bonsai	32.398	0.154	32.235	32.541	—
`3dgsmcmc`	Lego	39.649	0.071	39.597	39.730	—
`analyticsplatting`	Lego	36.797	1.333	35.994	38.335	Outlier
`erankgs`	Lego	16.878	1.504	15.573	18.522	Failure
`ges`	Lego	37.282	0.091	37.177	37.339	—
`lightgaussian`	Lego	36.133	0.079	36.042	36.189	—
`minisplatting`	Lego	38.673	0.008	38.666	38.682	—
`opti3dgs`	Lego	38.436	0.051	38.392	38.492	—
`pgsr`	Lego	32.789	0.004	32.785	32.793	—
`steepgs`	Lego	38.550	0.063	38.488	38.613	—
`vanilla_3dgs`	Lego	33.742	0.018	33.724	33.761	—

Cohort statistics. Across the 19 valid (method, scene) cells above, after excluding the `erankgs` × lego METHOD_FAILURE cell, the cohort PSNR-standard-deviation median is 0.058 dB and the P_{95} is 0.384 dB. The P_{95} value reported here and in section 3.3 is the conservative figure that includes the `analyticsplatting` × lego outlier ($\sigma = 1.333$ dB); excluding this annotated outlier yields a P_{95} of 0.173 dB. We adopt the conservative figure as the seed-noise floor in the main text. The `erankgs` × lego METHOD_FAILURE cell (mean PSNR 16.878 dB, range 2.949 dB across

Table 10: Per-scene Kendall τ over per-seed PSNR rankings, with permutation p-values, per seed pair.

Scene	Seed pair	τ	p
BONSAI	default vs. seed1	0.7778	0.0009
BONSAI	default vs. seed2	0.7333	0.0022
BONSAI	seed1 vs. seed2	0.9556	<0.001
BONSAI median τ		0.7778	—
LEGO	default vs. seed1	0.8571	0.0017
LEGO	default vs. seed2	0.8571	0.0017
LEGO	seed1 vs. seed2	1.0000	<0.001
LEGO median τ		0.8571	—

three failed reconstructions) is excluded from both figures.

Caveats on individual cells. Two cells in the seed-variance corpus carry annotations that recur in the rest of the paper.

> `analyticsplatting` exhibits anomalously high seed sensitivity on LEGO ($\sigma = 1.333$ dB, range = 2.341 dB; outlier flag set), compared to the cohort median of 0.058 dB. All three seeds produce valid reconstructions (PSNR = 38.33/36.06/35.99 dB); the variance likely reflects a sharp optimisation landscape introduced by the anti-aliasing rasteriser. `analyticsplatting` is retained in cohort statistics with outlier annotation.

> `erankgs` is excluded from seed-variance statistics for LEGO (STATUS = METHOD_FAILURE; mean PSNR 16.878 dB across all three seeds, range 2.949 dB). `erankgs` is retained in table 6 as a failure-mode exemplar and elsewhere in the benchmark tables.

E Method-by-Dataset Metric Grid

This appendix reports the cross-scene-mean PSNR, SSIM, and LPIPS for all $N_2 = 31$ methods on each of the four standard datasets used in SplatAtlas (Tanks & Temples, Mip-NeRF 360, NeRF-Synthetic, and Deep Blending), together with a per-scene weighted mean across all 31 scenes (weights 12/9/8/2 for T&T / Mip-NeRF 360 / NeRF-Synthetic / Deep Blending). Methods are grouped by their literature category (table 6) and ordered alphabetically within each group. The headline saturation and method-dominated representation findings of section 4 are computed on the per-scene values that underlie this grid; the values below are provided as a reference for between-method comparison at the dataset level.

E.1 Method-Level Failures and Reproduction Caveats

Under the unified protocol of section 3.1 (default hyperparameters, 30k steps), approximately 8% of (method, scene) cells in the $N_2 \times 31$ corpus exhibit method-level training anomalies. We document these here rather than silently excluding them, since the failure pattern itself is part of what a measurement study should report.

Training-stage failures. Several methods fail to converge on specific scenes (or whole datasets) under the default protocol. The most prominent cases include: (i) `eRankGS` × LEGO (PSNR ≈ 17 dB across all three seeds; section 3.3 and Appendix D); (ii) `GHAP` × NeRF-Synthetic (cross-scene mean PSNR 6.35 dB); (iii) `TrimGS` × Tanks & Temples (mean 16.44 dB); and (iv) `3DGS-MCMC` on T&T, NeRF-Synthetic, and Deep Blending (cross-scene mean PSNR 20.16/18.33/20.99 dB, against `vanilla_3dgs` 23.50/32.14/27.39 dB and the same method's strongest dataset-level result on Mip-NeRF 360 of 28.16 dB). Unlike (ii) and (iii), case (iv) is a partial-failure pattern: the seed-variance

entry on LEGO reaches 39.65 dB across all three seeds (table 9), so the NeRF-Synthetic dataset-level shortfall arises from the remaining seven scenes rather than from LEGO; back-of-envelope arithmetic gives a non-LEGO mean of roughly 15.3 dB. We do not claim definitive root-cause attribution for any of these failures; we did not succeed in recovering them under the default protocol. A plausible contributing factor in the NeRF-Synthetic cases (ii) and (iv) is that several methods rely on SfM-derived initial Gaussian positions; the synthetic Blender dataset provides no real SfM point cloud, and substituting random initialisation may produce qualitatively different optimisation trajectories. These failures are retained in tables 11 to 13 to reflect the actual behaviour of these methods under unified default settings.

Representation-level anomalies. `Gaussian Surfels` under default hyperparameters exhibits cross-scene weighted mean PSNR 15.64 dB, substantially below comparable surface-reconstruction methods (PGSR 26.40, 2DGS 27.74). A plausible reading is that surfel-based methods may require tuned per-scene hyperparameters that the unified protocol does not provide; we do not investigate this further and report the values as observed.

Implication for headline statistics. Section 4 reports ANOVA, CCF, and clustering results computed on all reproducible cells. We do not exclude failure cells from these statistics, since the failure pattern itself is part of method-protocol interaction; sensitivity analyses excluding the cells documented above are deferred to future work.

E.2 Cross-Scene Mean Tables

F Cell-Level SH–Loss Co-Variation

Section 4.2 reports that `sh_energy_ratio`, alone among the end-state attributes, behaves as a scene-dominated quantity (scene 57%, method 17%) within an otherwise method-dominated set. SH–geometry compensation is a natural conjecture for this exception: SH coefficients carry view-dependent appearance, and a model that misplaces geometry could in principle absorb the resulting error into more elaborate SH terms rather than into the loss. The simplest cell-level reading of this conjecture is that, across trained models, higher SH energy should accompany lower final loss—the compensation, if active at the cell level, would let methods buy lower loss with more SH.

We test this simplest reading on the 28-method trajectory cohort across the 31 canonical scenes (868 method–scene cells). Final SH mean and final training loss are positively associated, with a global Spearman correlation of $r = 0.417$ ($p = 8.20 \times 10^{-38}$). The pattern, were the cell-level compensation operating as conjectured, would yield a negative slope; the observed slope is positive. The direction is not a between-method artifact: per-method Spearman correlations have median 0.559 (IQR $[0.373, 0.706]$), and the sign is positive for 25 of the 28 methods. The three methods with negative per-method correlations (`trimgs` $r = -0.316$, `erankgs` $r = -0.083$, `coadaptgs` $r = -0.066$) all fail to reach significance at $n = 31$ per method ($p = 0.084$, 0.66, 0.72 respectively); we do not interpret them as evidence for compensation.

We read this as ruling out the simplest cell-level signature of SH–geometry compensation, not the conjecture in general. The conjecture admits readings at finer granularity—per-pixel, per-region, or conditional on local geometric error—that this end-state cell-level test does not address. What the cell-level data do show is co-variation rather than trade-off: SH energy and loss move together across scenes (with training loss as a difficulty proxy), consistent with both reflecting representational difficulty.

Table 11: PSNR (dB) across four datasets, cross-scene mean per (method, dataset). Higher is better. Source: aggregated from per-scene values in the SplatAtlas core bundle. Mean column is per-scene weighted across all 31 scenes (weights 12/9/8/2). †method-level training anomaly under the unified default-hyperparameter protocol; see Appendix E.1.

Method	T&T	Mip-NeRF 360	NeRF-Synthetic	Deep Blending	Mean
FREE					
3DGS-MCMC	20.16†	28.16	18.33†	20.99†	22.07
AbsGS	22.71	26.92	32.08	27.07	26.63
AtomGS	22.89	27.98	34.17	29.80	27.72
ConeGS	21.44	28.31	35.43	29.53	27.57
GES	23.25	27.26	34.62	27.28	27.61
GHAP	23.29	26.53	6.35†	28.21	20.17
GSLPM	23.41	27.80	35.21	27.09	27.97
LapisGS	22.81	27.27	35.28	24.31	27.42
ReactGS	22.80	27.76	35.38	26.00	27.69
3DGS	23.50	27.79	32.14	27.39	27.23
GEO_REG					
2DGS	22.99	27.62	35.14	27.09	27.74
Gaussian Surfels	14.43	13.52	20.73	12.07	15.64†
GOF	22.73	25.37	32.68	29.49	26.50
PGSR	22.43	27.12	31.40	27.07	26.40
Scaffold-GS	19.49	24.23	26.52	25.74	23.09
PRIM_CTRL					
CoAdaptGS	16.86	21.29	22.57	18.60	19.73
eRankGS	21.08	26.61	17.68	25.77	22.11
GaussianPro	23.27	27.70	35.56	26.87	27.96
MiniSplatting	22.66	27.55	34.48	27.55	27.45
Opti3DGS	23.38	27.65	35.11	27.41	27.91
SteepGS	23.25	27.66	35.15	27.04	27.84
RENDER					
3DGS-DR	21.79	26.81	34.57	25.98	26.81
AnalyticSplatting	23.38	27.82	35.17	27.11	27.95
LoD-GS	25.41	29.00	38.21	28.32	29.94
Mip-Splatting	22.77	27.92	35.60	26.26	27.80
PixelGS	23.27	27.84	35.49	26.37	27.95
COMPRESS					
CDCGS	23.11	28.62	31.59	31.40	27.44
HoGS	23.53	27.83	35.44	26.40	28.04
LightGaussian	22.41	27.35	33.43	25.12	26.86
Octree-GS	20.37	25.30	33.06	27.30	25.53
TrimGS	16.44†	28.20	33.67	22.56	24.69

G Hypothesis Mapping from Counterfactual Channels to Design Directions

Table 14 translates channel-level signals into candidate design directions, with each row backed by a paired comparison from section 5.6. We present this as a preliminary hypothesis mapping rather than a set of proven recipes; the two unanimous pairs supply the strongest support, the SteepGS row is directionally consistent at 0.839, and the LightGaussian row is flagged as a warning rather

Table 12: SSIM across four datasets, cross-scene mean per (method, dataset). Higher is better. †method-level training anomaly under the unified default-hyperparameter protocol; see Appendix E.1.

Method	T&T	Mip-NeRF 360	NeRF-Synthetic	Deep Blending	Mean
FREE					
3DGS-MCMC	0.7280†	0.8428	0.7635†	0.7237†	0.7702
AbsGS	0.7936	0.7989	0.9666	0.8090	0.8408
AtomGS	0.7898	0.8135	0.9709	0.8338	0.8463
ConeGS	0.7931	0.8351	0.9757	0.8238	0.8544
GES	0.8169	0.8051	0.9730	0.8238	0.8542
GHAP	0.8095	0.7683	0.2380†	0.6921	0.6425
GSLPM	0.8203	0.8299	0.9757	0.8205	0.8632
LapisGS	0.8234	0.8213	0.9774	0.7674	0.8589
ReactGS	0.8110	0.8345	0.9761	0.8019	0.8598
3DGS	0.8253	0.8256	0.9663	0.8237	0.8617
GEO_REG					
2DGS	0.8126	0.8184	0.9749	0.8165	0.8564
Gaussian Surfels	0.5662	0.4279	0.8682	0.6458	0.6091†
GOF	0.8088	0.7672	0.9782	0.9047	0.8466
PGSR	0.8014	0.8248	0.9623	0.8143	0.8506
Scaffold-GS	0.7215	0.7015	0.9044	0.7984	0.7678
PRIM_CTRL					
CoAdaptGS	0.5527	0.5666	0.8379	0.7007	0.6399
eRankGS	0.7658	0.8115	0.7993	0.8093	0.7905
GaussianPro	0.8159	0.8223	0.9771	0.8155	0.8594
MiniSplatting	0.8079	0.8324	0.9736	0.8293	0.8592
Opti3DGS	0.8204	0.8266	0.9751	0.8244	0.8624
SteepGS	0.8169	0.8205	0.9751	0.8180	0.8588
RENDER					
3DGS-DR	0.7758	0.8001	0.9720	0.8115	0.8358
AnalyticSplatting	0.8217	0.8261	0.9751	0.8196	0.8624
LoD-GS	0.9035	0.8642	0.9911	0.7892	0.9073
Mip-Splatting	0.8136	0.8365	0.9770	0.8003	0.8615
PixelGS	0.8220	0.8350	0.9765	0.8043	0.8645
COMPRESS					
CDCGS	0.8125	0.8515	0.9659	0.9197	0.8703
HoGS	0.8227	0.8300	0.9756	0.8096	0.8634
LightGaussian	0.7858	0.8133	0.9684	0.8005	0.8419
Octree-GS	0.7761	0.7712	0.9674	0.8177	0.8267
TrimGS	0.5617†	0.8404	0.9719	0.7175	0.7585

than a prescription.

H Counterfactual Sensitivity on the Comparable Subset of N_3

The dispersion analysis of section 5.4 restricts to the $k = 5$ PSNR-saturated cluster (approximately 1.5 dB). For context, the following values report the same six counterfactual channels computed across the comparable subset of N_3 (8 methods, excluding AnalyticSplatting), which permits PSNR differences up to several dB: $\rho_{\text{opa}} = 55.67\%$ (8.6× the PSNR relative IQR), $\phi_{\text{cov}} = 35.86\%$ (5.6×), $\Delta\varepsilon_{\text{opa}} = 25.36\%$ (3.9×), $\Delta\varepsilon_{\text{SH}} = 23.97\%$ (3.7×). Two channels, ρ_{SH} (5.74%) and ε_{res} (6.04%), fall within the PSNR range and provide no additional method-level discrimination at either scope. This

Table 13: LPIPS across four datasets, cross-scene mean per (method, dataset). Lower is better. †method-level training anomaly under the unified default-hyperparameter protocol; see Appendix E.1.

Method	T&T	Mip-NeRF 360	NeRF-Synthetic	Deep Blending	Mean
FREE					
3DGS-MCMC	0.3543†	0.2141	0.2223†	0.5066†	0.2893
AbsGS	0.2611	0.2483	0.0578	0.3574	0.2111
AtomGS	0.2772	0.2404	0.0465	0.3445	0.2113
ConeGS	0.3334	0.2290	0.0352	0.3562	0.2276
GES	0.2579	0.2819	0.0391	0.3542	0.2146
GHAP	0.2952	0.3519	0.4797†	0.4791	0.3711
GSLPM	0.2512	0.2431	0.0343	0.3549	0.1996
LapisGS	0.2181	0.2368	0.0306	0.4107	0.1876
ReactGS	0.2462	0.2122	0.0335	0.3774	0.1899
3DGS	0.2359	0.2430	0.0603	0.3452	0.1997
GEO_REG					
2DGS	0.2570	0.2520	0.0353	0.3601	0.2050
Gaussian Surfels	0.4563	0.5042	0.1225	0.5069	0.3873†
GOF	0.2626	0.2564	0.0315	0.2746	0.2019
PGSR	0.2719	0.2513	0.0650	0.3694	0.2188
Scaffold-GS	0.3645	0.4221	0.1355	0.4495	0.3276
PRIM_CTRL					
CoAdaptGS	0.5850	0.5410	0.1711	0.5847	0.4654
eRankGS	0.3166	0.2472	0.1887	0.3890	0.2681
GaussianPro	0.2538	0.2478	0.0311	0.3617	0.2015
MiniSplatting	0.2739	0.2453	0.0358	0.3563	0.2095
Opti3DGS	0.2486	0.2481	0.0353	0.3471	0.1997
SteepGS	0.2508	0.2476	0.0353	0.3572	0.2011
RENDER					
3DGS-DR	0.3189	0.2892	0.0400	0.3916	0.2430
AnalyticSplatting	0.2457	0.2449	0.0358	0.3540	0.1983
LoD-GS	0.1098	0.1646	0.0086	0.1700	0.1035
Mip-Splatting	0.2422	0.2084	0.0319	0.3701	0.1864
PixelGS	0.2295	0.2079	0.0323	0.3676	0.1812
COMPRESS					
CDCGS	0.2647	0.2198	0.0588	0.2757	0.1992
HoGS	0.2356	0.2260	0.0343	0.3619	0.1890
LightGaussian	0.3160	0.2746	0.0479	0.4052	0.2405
Octree-GS	0.2456	0.2793	0.0462	0.3521	0.2108
TrimGS	0.5264†	0.2248	0.0415	0.4345	0.3078

wider-subset sensitivity check preserves the qualitative finding that several counterfactual channels show greater relative dispersion than PSNR, although the exact channel ordering differs from the $k = 5$ saturated-cluster ordering.

I Metric PCA

A principal component analysis of the three metrics (LPIPS sign-flipped, then standardised) yields PC1 accounting for 88.4% of total variance with near-equal loadings on PSNR (0.575), SSIM (0.590), and LPIPS (0.567); PC2 accounts for 7.8% and is LPIPS-dominated (loading 0.772). Within the saturated regime, the three indicators are nearly interchangeable as scalar summaries: LPIPS

Table 14: Hypothesis mapping from diagnostic signals to design directions. "Support" references pairs in table 4.

Diagnostic signal	Suggested direction	Support	Confidence
High $\Delta\varepsilon_{\text{opa}}$ (> 0.10)	Geometric regularisation (PGSR-style planar constraint)	P01 (agr. 1.000)	High
High ϕ_{cov} (> 0.10)	Rank-aware primitive control (eRankGS-style)	P02 (agr. 1.000)	High
Moderate $\Delta\varepsilon_{\text{opa}}$ (0.05–0.10), steep geometry	Steepness-aware densification (SteepGS-style)	P04 (agr. 0.839)	Medium
High ρ_{opa} in compression-class methods	*Warning*: pruning may aggravate pathology rate—trade-off with compression ratio	P03 (reverse)	Medium (reverse)

provides a supplementary rather than qualitatively distinct signal.

J Mechanism Scope and Late-Stage Freeze

Two further observations sharpen the dynamics picture along orthogonal axes. *Mechanism scope.* Densification events selectively perturb three of four dimensions: opacity, scale, and *sh_dc_mag* all show medium-to-large effect sizes (0.53–0.56 at $|Z| > 87$), whereas anisotropy shows a near-zero effect of opposite sign (effect size -0.053, $Z = -8.80$; fig. 4C). The dominant primitive-control mechanism does not act uniformly across the representational dimensions it is widely assumed to shape. *Late-stage freeze.* Once the population stabilises, structural change largely ceases: 77% (24/31) of methods show stable-phase growth rate exactly zero in steps 15,000–30,000. Restricting to the trajectory sub-cohort, the adaptability index has median 0.800 with 96.4% (27/28) of methods below 1.0 (fig. 4B), indicating that the frozen structure is method-determined rather than scene-driven. The sole exception is `3dgsmcmc` (adaptability 1.012); we flag this rather than interpret it.

K Literature Taxonomy: Detailed Cluster Analysis

The data-driven reclassification does not invalidate the taxonomy. Taxonomies classify by design intent; trajectory clusters classify by evolutionary path. The two need not align in principle.

A specific structural observation: three of the four data-driven clusters are dominated by PRIM_CTRL methods, and 16 of the 17 reclassified methods are assigned to PRIM_CTRL-majority clusters. We report this pattern descriptively; mechanistic interpretation is not separable from feature-selection considerations under the present feature set.

L Saturated Cluster Membership

The five methods comprising the saturated cluster (approximately 1.5 dB of the top performer) are: `vanilla_3dgs` Kerbl et al. [2023], `pgsr` Chen et al. [2025a], `ges` Hamdi et al. [2024], `opti3dgs` Farooq et al. [2025], and `steepgs` Wang et al. [2025a]. Within this cluster, point estimates are descriptive rather than inferential; we report dispersion ratios as effect sizes and do not assert population-level significance.

M Threshold Sweep for eRankGS Direction Agreement

Table 15 reports the P02 direction agreement (eRankGS vs. vanilla-3DGS on ϕ_{cov}) under progressive exclusion of scenes where eRankGS reconstruction quality falls below a PSNR threshold t. Direction agreement remains 1.000 at all thresholds, confirming that the validation result is not inflated by

catastrophically failed reconstructions.

Table 15: P02 direction agreement under PSNR-threshold exclusion.

Threshold t (dB)	Scenes retained	Scenes with $\Delta\phi_{\text{cov}} < 0$	Direction agreement
(none)	31	31	1.000
≥ 17	20	20	1.000
≥ 19	20	20	1.000
≥ 21	17	17	1.000
≥ 23	10	10	1.000
≥ 25	8	8	1.000